
Internet of Things (IoT)

— Chapter 4 —

Learning outcomes

After completing this session, you should be able to

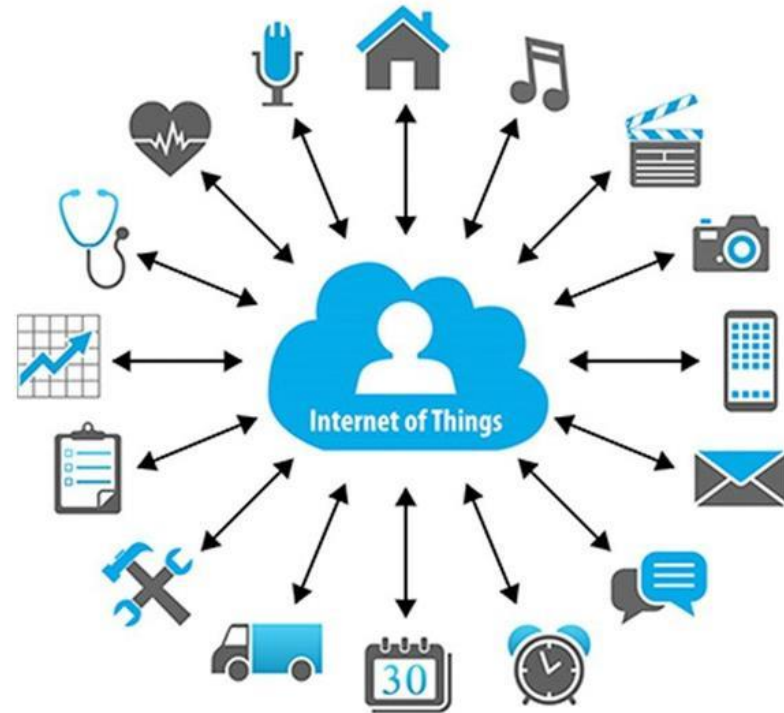
Describe IoT

Describe the pros and cons of IoT

Explain how IoT works

Explain the architecture of IoT

Describe application areas of IoT



IAB's Definition of IoT

According to the Internet Architecture Board's (IAB) definition,

IoT is the **networking of smart objects**, meaning a huge number of devices **intelligently communicating** in the presence of internet protocol that cannot be directly operated by human beings but **exist as components** in buildings, vehicles or the environment.

IETF's Definition of IoT

According to the Internet Engineering Task Force (IETF) organization's definition,

IoT is the **networking of smart objects** in which smart objects **have some constraints** such as limited bandwidth, power, and processing accessibility for achieving interoperability among smart objects

Definition of IoT

The Internet of Things (IoT) is

the **network of physical objects** or "things" **embedded** with electronics, software, sensors, and network connectivity, which **enables** these **objects** to **collect** and **exchange** data

Simply stated, the Internet of Things consists of **any device with an on/off switch** connected to the Internet

Overview of IoT

The most important features of IoT include

Artificial Intelligence

To make virtually anything “smart”

Connectivity

To create small networks between its system devices

Sensors

defining instruments that makes IoT device an active system capable of real-world integration

Overview of IoT

The most important features of IoT include

Active engagement

IoT introduces a new paradigm for active content, product, or service engagement

Small Device

IoT exploits **purpose-built small devices** to deliver its **precision, scalability, and versatility**

IoT – Advantages

Improved Customer Engagement

IoT enables to achieve richer and more effective engagement with audiences

Technology Optimization

IoT can be used to improve device use, and to aid in more potent improvements to technology

Reduced Waste

IoT provides real-world information leading to the more effective management of resources

IoT – Advantages

Enhanced Data Collection

It allows an accurate picture of everything through data

IoT – Disadvantages

The more information is shared between devices, the more **potential that a hacker could steal confidential information**

If there's a bug in the system, it's likely that every **connected device will become corrupted**

IoT device **incompatibility** due to lack of international standard

Challenge in collecting and managing data from huge number of devices

Challenges of IoT

Security

constantly connected devices communicating over networks exposes users to various kinds of security attack

Privacy

substantial personal data in extreme detail might leak without even the active participation of users

Challenges of IoT

Complexity

- in terms of design, deployment, and maintenance
- use of multiple technologies and a large set of new enabling technologies

Compliance

Its complexity makes the issue of compliance seem incredibly challenging

How IoT work?

An IoT ecosystem consists of **web-enabled smart devices**

These devices **collect, send** and **act on** data they acquire from their environments using their

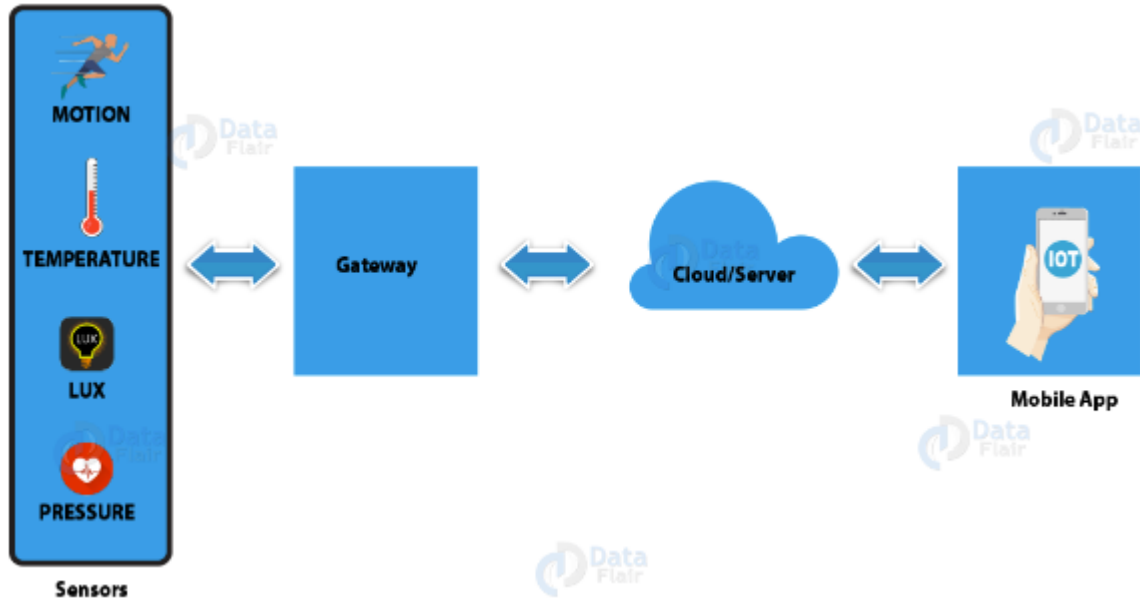
sensors,

embedded processors, and

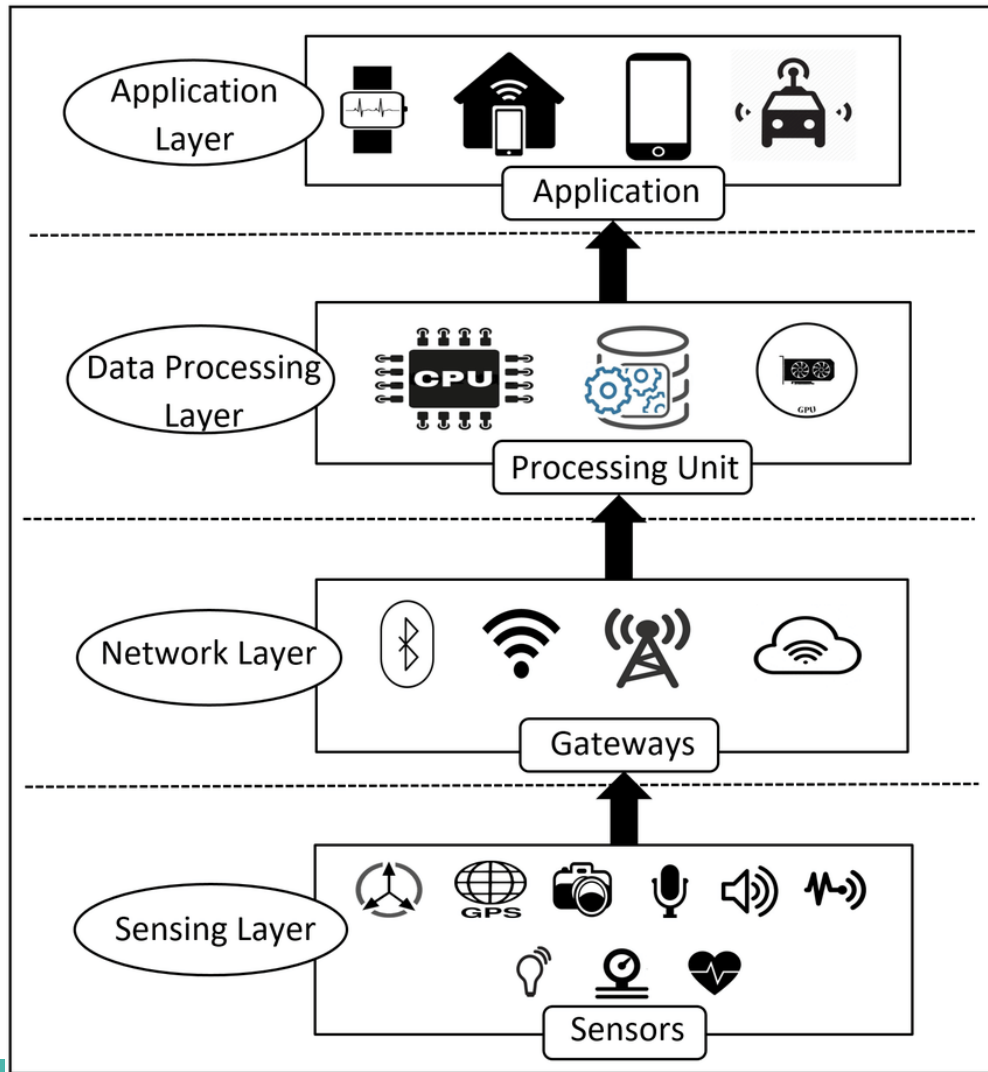
communication hardware

How IoT work?

IoT devices share the sensor data they collect by connecting to an **IoT gateway** or another **edge device** where data is either sent to the **cloud** to be analyzed or analyzed locally

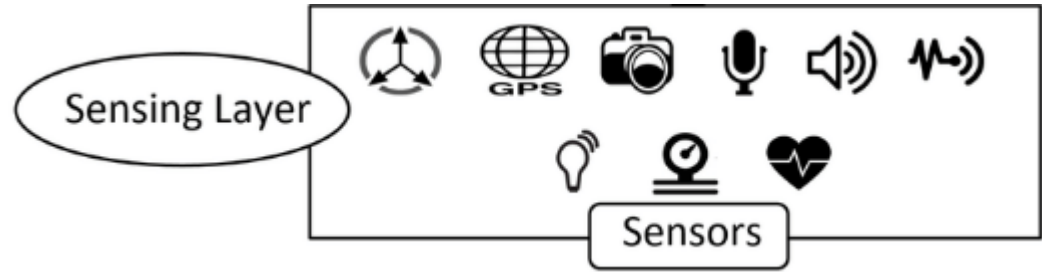


Architecture of IoT



Architecture of IoT

Sensing Layer



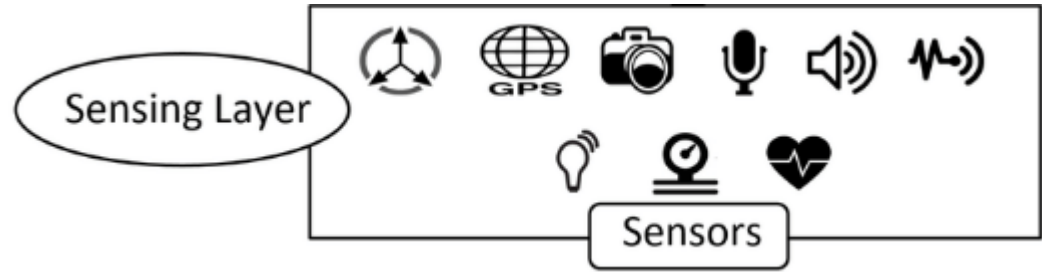
identify any phenomena in the devices' peripheral and obtain data from the real world

Sensors in IoT devices are usually integrated through **sensor hubs**

A **sensor hub** is a common connection point for multiple sensors that accumulate and forward sensor data to the processing unit of a device

Architecture of IoT

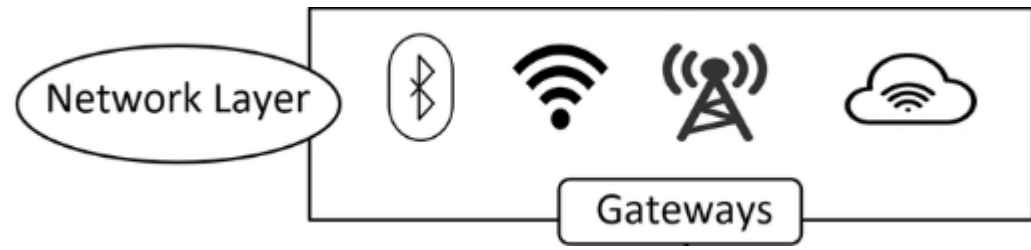
Sensing Layer



Sensors in IoT devices can be classified into three broad categories

1. **Motion Sensors:** measure the change in motion (angular or linear) as well as the orientation of the devices
2. **Environmental Sensors:** sense the change in environmental parameters in the device's peripheral. E.g. Light and Pressure sensors
3. **Position sensors:** deal with the physical position and location of the device. E.g. Global Positioning System (GPS) and Magnetic Sensors

Architecture of IoT



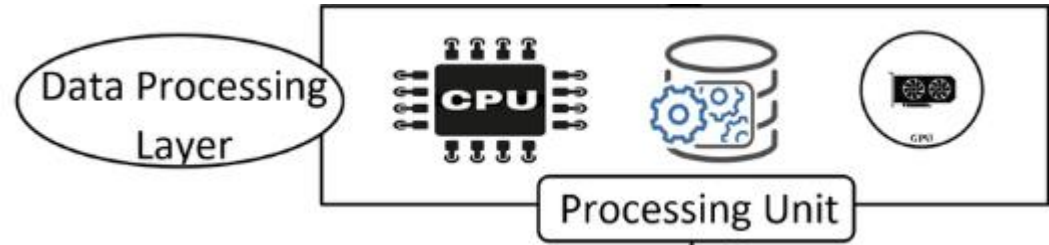
Network Layer

acts as a communication channel to transfer data, collected in the sensing layer, to other connected devices

this layer is implemented by using diverse communication technologies such as

Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRa, Cellular Network

Architecture of IoT



Data Processing Layer

takes data collected in the sensing layer and **analyses the data to make decisions** based on the result

in some IoT devices (e.g., Smart Watch and Smart Home Hub), the data processing layer **also saves the result** of the previous analysis to improve the user experience

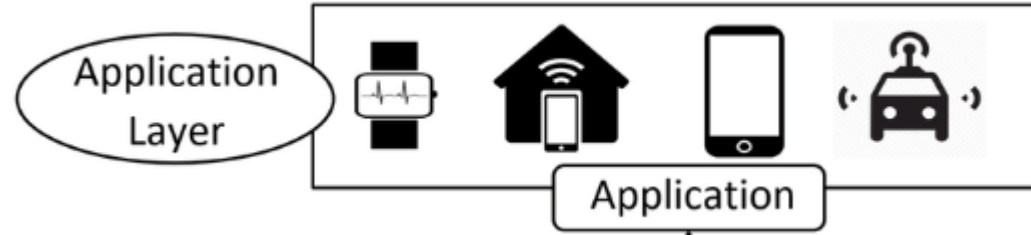
this layer **may share the result of data processing** with other connected devices via the network layer

Architecture of IoT

Application Layer

Implements and presents the results of the data processing layer to accomplish disparate applications of IoT devices

Is a user-centric layer that executes various tasks for the users



Devices and Networks

Connected IoT devices can be categorized into three main groups

Consumer, Enterprise and Industrial

Consumer connected devices include

Smart TVs, Smart Speakers, Toys, Wearables, and Smart Appliances

Enterprise and Industrial connected devices include

Smart Meters, Commercial Security Systems and Smart City technologies

Devices and Networks

Other technologies such as

Smart Air Conditioning, Smart Thermostats, Smart Lighting, and Smart Security, span consumer, enterprise, and industrial uses

Devices and Networks

IoT network typically includes a number of devices with constrained resources such as power, processing, and memory

The **efficient management** of IoT networks requires **considering** both the **constraints** of low power IoT devices and the **deployment complexity** of the underlying communication infrastructure

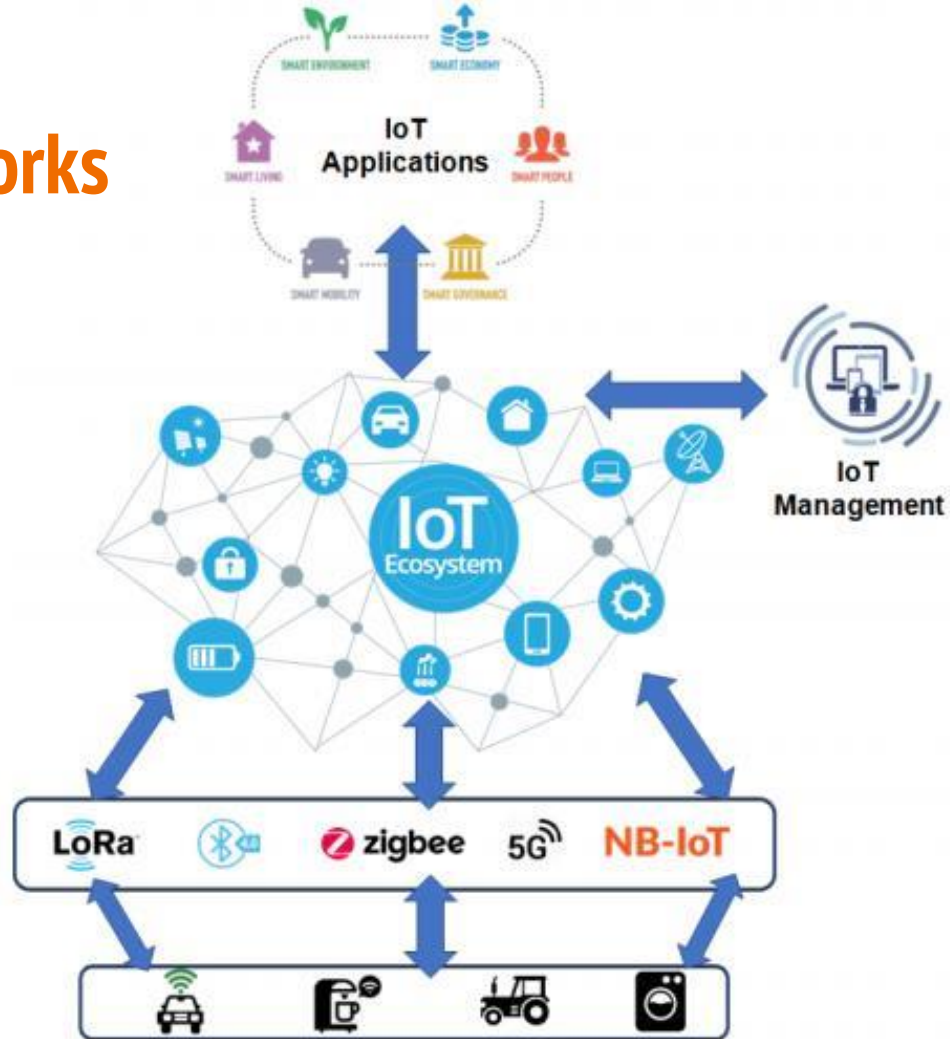
To ensure the correct functioning of those connected devices, they must be **remotely accessed to configure** or **monitoring** their status

Devices and Networks

Efficient and autonomic management of IoT networks take into account several elements such as

Scalability, Interoperability, Energy Efficiency, Topology Control, Quality of Service (QoS), Fault Tolerance, and Security

Devices and Networks

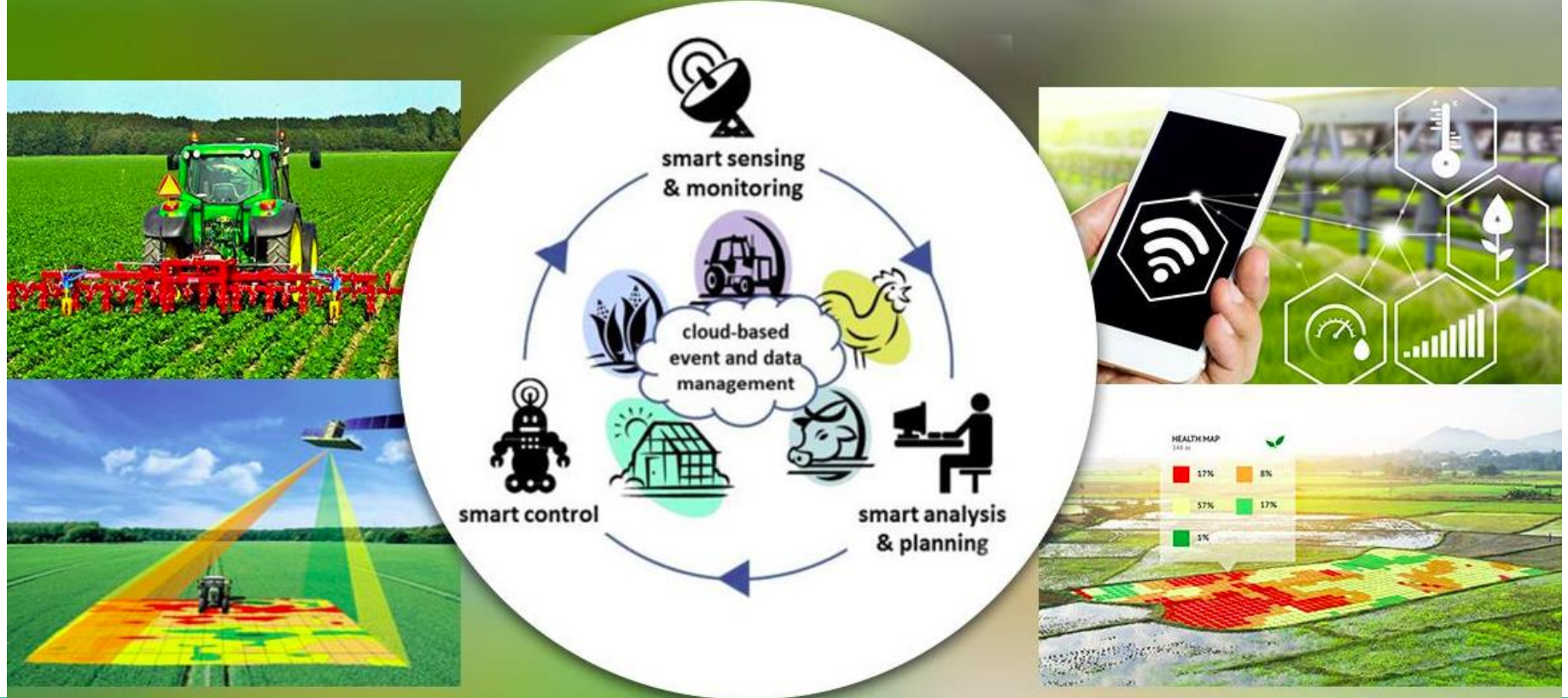


Applications of IoT: Agriculture

Monitoring and management of microclimate conditions for indoor planting

Sense soil moisture and nutrients, in conjunction with weather data, better control smart irrigation and fertilizer systems

Applications of IoT: Agriculture



Applications of IoT: Consumer Use

IoT devices such as Fitbit, Smart Phones, Apple Watches, Health Monitors help improve **entertainment, health, and fitness**



Applications of IoT: Consumer Use

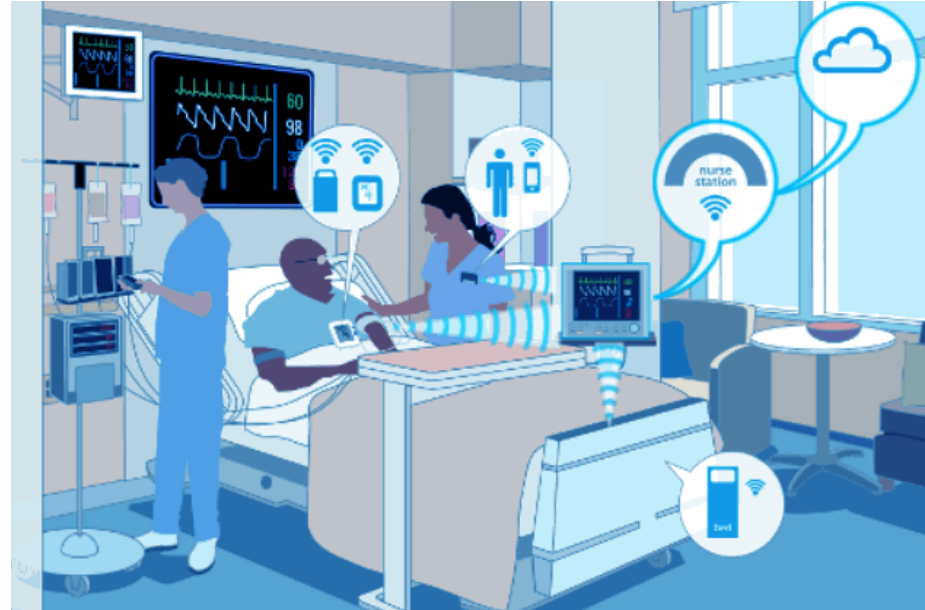
Smart homes take care of things like activating environmental controls so that your house is at peak comfort when you come home

Appliances such as Oven, Lights, and security devices such as Smart Locks can be controlled remotely



Applications of IoT: Healthcare

Help monitor patients' health at home, thereby reducing hospital stays while still providing up to the minute real-time information that could save lives



Applications of IoT: Manufacturing

RFID and GPS technology can help a manufacturer track a product from its start on the factory floor to its placement in the destination store, the whole **supply chain** from start to finish

Sensors can **gather** information on **travel time**, **product condition**, and **environmental conditions** that the product was subjected to



Applications of IoT: Retail

IoT can help analyze mall traffic to enhance the customer's shopping experience

IoT helps retailers target customers (for instance for advertisement) based on past purchases

Automated checkout,
store layout optimization,
Smart shelves, Beacons



Applications of IoT: Transportation

IoT for Self-Driving cars

GPS data can be used to determine traffic patterns, parking space demand, and road construction and maintenance



Applications of IoT: Utilities

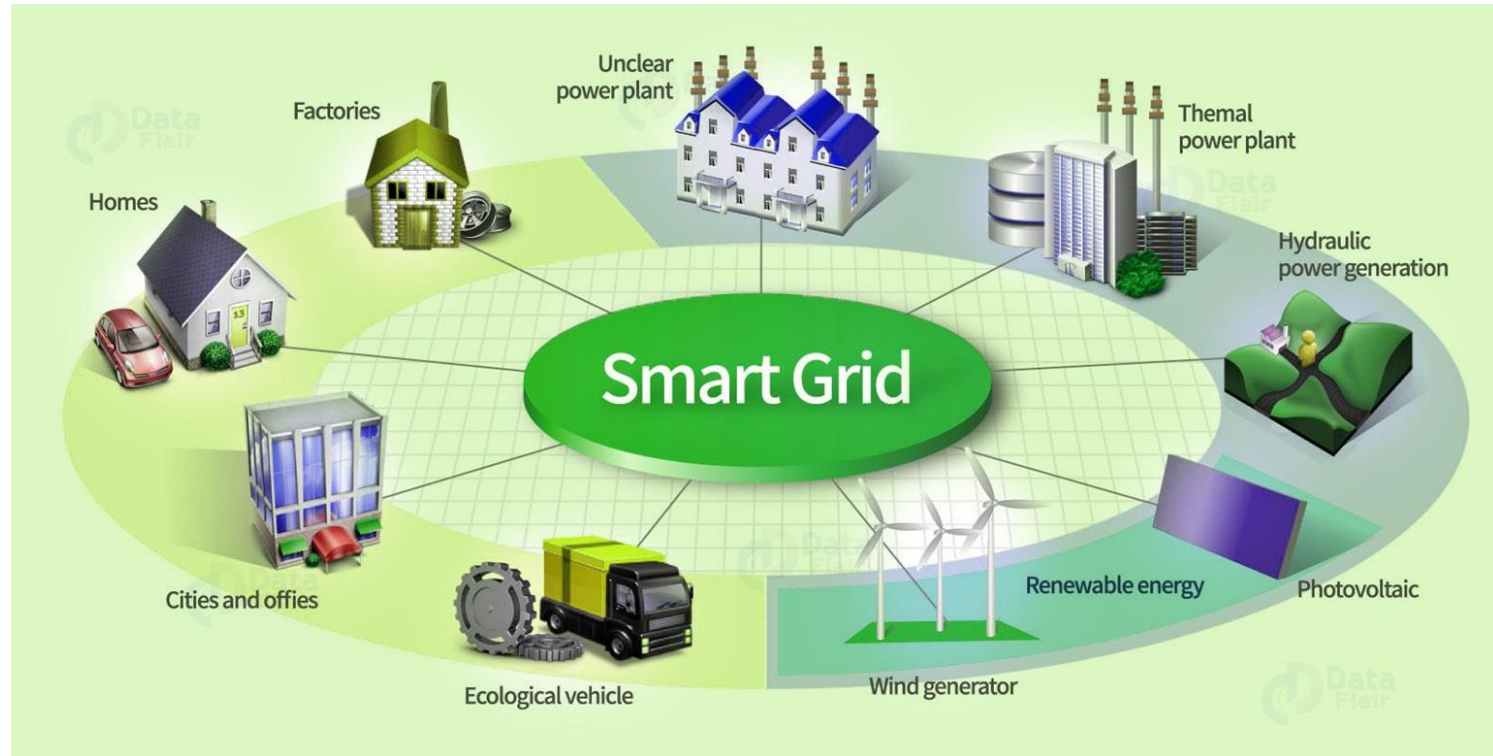
IoT sensors can be employed to monitor environmental conditions such as **humidity, temperature, and lighting**

With IoT-driven environmental control, businesses and private residences can experience significant **energy savings**

Data gathered by the Internet of Things can be used to help run municipal power grids more efficiently, analyzing factors such as usage

Sensors can help pinpoint outages faster

Applications of IoT: Utilities



Applications of IoT: Utilities

